

# DD-Mamba: Dual-Domain State-Space Forecasting with a Linear Time Backbone and a Conditional Spectral Pathway

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**Abstract.** Long-term multivariate forecasting must reconcile local time-domain dynamics with periodic structure represented compactly in the frequency domain. We present DD-Mamba, a dual-domain framework whose decomposition-based time branch and learned spectral branch each produce a forecast, fused by a convex gate. Two empirical guidelines organize the design. First, a DLinear-style time-domain backbone is trained alongside zero-initialized nonlinear corrections; the optional FITS-style spectral map also starts dormant, so the initial prediction is a linear map of the instance-normalized window. Second, state-space capacity is placed per dataset: a bidirectional Mamba over variates is enabled where cross-channel structure supports it and disabled where it overfits. Across nine look-back-96 benchmarks and three seeds, DD-Mamba improves over reported baselines by more than the 0.005 descriptive tolerance on Solar, ETTh1, and ETTh2 — though only the Solar and ETTh1 margins also clear the wider 0.01 cross-paper caution — lies within that tolerance of the leading reported results on ETTm1, ETTm2, Weather, and Electricity, and trails on Traffic and Exchange. Because baseline numbers are quoted rather than uniformly re-run, these are cross-paper comparisons. Paired ablations at  $H=96$  show the variate mixer helps Weather yet hurts ETTh1, and RevIN helps Weather yet hurts Solar — capacity and normalization must be *placed*, not defaulted. A nine-dataset branch-ablation matrix finds the frequency path significant on four datasets (Solar, ETTm1, Electricity, Exchange), both branches jointly necessary on three, and the branches redundant elsewhere; zero initialization is accuracy-neutral. The evidence establishes conditional component placement: dual-domain value is real but dataset-conditional.

**Keywords:** Time series forecasting · State-space models · Mamba · Frequency domain · Multivariate analysis

## 1 Introduction

Long-term forecasting of multivariate time series underpins applications from power-grid load planning to traffic management and weather-dependent operations. The field has cycled through architectural paradigms at a remarkable pace:

Transformer variants with efficient attention [27, 22, 28], patch-based attention [17], inverted variate-token attention [13], and, most recently, selective state-space models (SSMs) built on Mamba [8, 20], which offer linear-time sequence modeling with input-dependent dynamics.

*Three empirical gaps* complicate this progress. **Gap 1: deep models discard strong linear maps.** A per-component linear projection from the look-back window to the horizon (DLinear [25]) and a tiny complex-linear interpolation of the low-passed spectrum (FITS [23]) can be competitive with much larger models on standard benchmarks. Without such a direct pathway, a deep architecture must learn any linear behavior implicitly, which may be inefficient on small datasets. **Gap 2: the useful inductive bias differs by domain.** Local trends and regime changes are naturally expressed in the time domain, while multi-scale seasonality is compact in the frequency domain, yet models committed to a single view — attention or SSMs over time steps only, or spectral mixing only [24] — do not expose both views as separately testable forecast paths. **Gap 3: cross-variate capacity can help or hurt.** Cross-variate mixing is effective on several high-dimensional benchmarks [13, 20], but our controlled results show that the same module can overfit a low-channel dataset. This motivates testing the mixer as an explicit option rather than assuming one setting is universal.

We answer these three gaps with DD-Mamba (Fig. 1), mapping each to one structural remedy: *(i)* a DLinear-style time backbone with zero-initialized nonlinear corrections; *(ii)* parallel time and frequency forecasts, including an optional dormant FITS-style spectral map, fused by a convex gate; and *(iii)* a bidirectional variate mixer switched on or off per dataset. Paired ablations test these choices below; only effects whose confidence intervals exclude zero are treated as supported.

Concretely, the time branch combines a DLinear-style pathway [25] with a zero-initialized nonlinear correction, while the frequency branch combines learned filtering with an optional zero-initialized FITS-style spectral pathway [23]. The resulting model starts as a linear map of the instance-normalized window (the RLinear-class structure [11, 10]); a controlled test finds zero initialization accuracy-neutral. The branch forecasts form a learned convex combination. The strongest ablation evidence concerns *placement*: removing the variate mixer hurts Weather whereas adding it hurts ETTh1, each with a paired CI excluding zero, and normalization shows the same two-sided pattern. A nine-dataset branch-ablation matrix then locates where each domain matters: the frequency path is individually significant on four datasets (Solar, ETTh1, Electricity, Exchange), both branches are jointly necessary on three, and the branches are redundant on ETTh1, ETTh2, Weather, and Traffic — so the dual-domain claim is supported, but explicitly dataset-conditional.

**Contributions. (1)** Two-sided, cross-dataset evidence that state-space capacity and normalization choices can help or hurt — the case for per-dataset placement over universal defaults; **(2)** a DLinear-style time backbone with zero-initialized deep corrections and an optional dormant spectral pathway, with a controlled

initialization ablation; **(3)** a dual-domain architecture whose output is a gated convex combination of independently testable branch forecasts, supported by a nine-dataset branch-ablation matrix that finds the frequency path significant on four datasets and redundant on the rest; and **(4)** three-seed results on nine benchmarks plus paired ablations, including statistically supported negative and dataset-dependent findings.

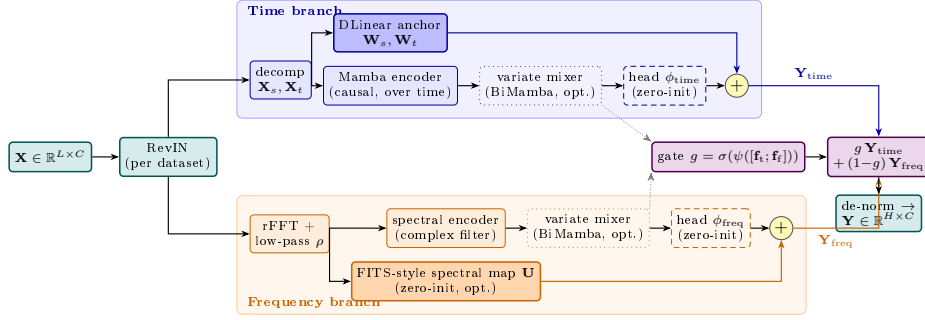
## 2 Related Work

*Transformer forecasters.* Informer [27], Autoformer [22] and FEDformer [28] adapt attention to long sequences via sparsity, decomposition, and frequency mixing; Non-stationary Transformers [14] restore the stationarity that normalization removes, and Crossformer [26] attends across both time and variates. PatchTST [17] treats sub-series patches as tokens with channel independence; iTransformer [13] inverts the axes and attends over *variates*, establishing that cross-channel modeling drives accuracy on high-dimensional benchmarks. SOFTS [9] uses a centralized series core for linear-complexity channel interaction, while ModernTCN [15] combines temporal and cross-variable convolutions. DD-Mamba adopts the variate-mixing insight with bidirectional SSMs, treats that mixer as optional, and couples it to an explicit frequency branch and a linear time backbone.

*Linear and frequency-domain models.* DLinear/NLinear [25] showed decomposition plus a linear map rivals Transformers; RLinear [11] added reversible instance normalization [10], and the MLP-based TiDE [6] and TimeMixer [19] confirm that lightweight mixers stay competitive. N-BEATS [18] and N-HiTS [3] provide the closest residual-forecasting precedents: they build forecasts through successive residual blocks, whereas DD-Mamba retains a direct time-domain linear forecast and trains selective-SSM features as a parallel correction. In the spectrum, FreTS [24] learns frequency-domain MLPs and FITS [23] forecasts with a single complex-linear interpolation of the low-passed spectrum at negligible cost. The DD-Mamba hypothesis class includes related linear configurations, but joint optimization does not guarantee recovering the corresponding standalone models.

*State-space models.* Mamba [8] introduced selective SSMs with input-dependent dynamics and linear-time scans, extended by Mamba-2 [5]. In forecasting, S-Mamba [20] and TimeMachine [1] apply bidirectional Mamba over variate tokens, and further works explore frequency-domain SSMs [16]. DD-Mamba unifies these threads: causal Mamba over time, bidirectional Mamba over frequency bins and variates, placed per dataset rather than uniformly.

*Zero-initialized corrections.* Zero-initialized residual paths are established optimization devices; for example, ReZero [2] learns a zero-initialized residual gate to improve signal propagation in very deep networks. Our use is narrower and



**Fig. 1.** DD-Mamba overview, color-coded by role: the **normalization** wrapper (teal), the **time branch** (blue), the **frequency branch** (orange), and the **convex-gate fusion** (violet). In each time branch, a solid-filled DLinear-style backbone runs in parallel with a zero-initialized nonlinear head (dashed). The optional frequency-domain map and its nonlinear head are also zero-initialized, so the model begins as a scaled linear map of the instance-normalized window. Dotted boxes are per-dataset switches; dotted grey arrows carry branch features to the gate, whose convex weight  $g$  mixes the two coloured forecast paths  $\mathbf{Y}_{\text{time}}$  and  $\mathbf{Y}_{\text{freq}}$ .

forecasting-specific: a complete linear forecast remains available while nonlinear time- and frequency-domain heads start silent. Because the controlled comparison is accuracy-neutral, we treat this as an interpretable initialization convention, not a new optimization theorem or an accuracy contribution.

*Decomposition and heteroscedasticity.* Splitting a series into interpretable components is a century-old idea: classical additive models, STL [4], wavelet analysis, EMD, and SSA extract a trend, a seasonal part, and an irregular remainder, and learned decompositions such as SCINet [12] carry it into deep forecasters. A recurring limitation, made explicit by the seasonal-trend-*dispersion* decomposition (STD) [7], is that time-varying scale can remain entangled with seasonal or irregular components. RevIN [10], used in DD-Mamba as a per-dataset option, is related only at the level of motivation: it reversibly normalizes each input window by its observed mean and standard deviation, but neither extracts an interpretable dispersion component nor forecasts future scale. We therefore do not claim an STD/STDR-style decomposition. Instead, the present experiments test the narrower question of when window-level normalization helps or hurts, while explicit dispersion forecasting remains future work.

### 3 Method

#### 3.1 Problem Formulation and Normalization

Given a look-back window  $\mathbf{X} \in \mathbb{R}^{L \times C}$  of  $L$  steps and  $C$  variates, the task is to predict the next  $H$  steps  $\mathbf{Y} \in \mathbb{R}^{H \times C}$ . Following RevIN [10], we standardize

each window per variate,  $\tilde{\mathbf{X}} = (\mathbf{X} - \boldsymbol{\mu})/\boldsymbol{\sigma}$ , and de-normalize the final forecast. Instance normalization is treated as a per-dataset switch rather than a universal default; Sect. 4.3 shows why.

### 3.2 Time Branch: Decomposition, Linear Backbone, Selective Scan

The time branch decomposes  $\tilde{\mathbf{X}}$  with a moving average into trend  $\mathbf{X}_t$  and seasonal  $\mathbf{X}_s$  components [22, 25] and forecasts with two cooperating paths.

*Linear backbone.* Per-component linear maps project the window directly to the horizon:

$$\mathbf{Y}_{\text{lin}} = \mathbf{W}_s \mathbf{X}_s + \mathbf{W}_t \mathbf{X}_t \in \mathbb{R}^{H \times C}, \quad \mathbf{W}_s, \mathbf{W}_t \in \mathbb{R}^{H \times L}, \quad (1)$$

i.e., a DLinear-style map [25] (randomly initialized and trained jointly — a parameterized pathway, not a fitted DLinear), shared across variates.

*Selective-scan correction.* The pair  $[\mathbf{X}_s; \mathbf{X}_t]$  is embedded per time step and processed channel-independently by a stack of Mamba layers [8] (Fig. 2a). A Mamba layer discretizes a state-space system with input-dependent parameters  $(\boldsymbol{\Delta}, \mathbf{B}, \mathbf{C})$ :

$$\mathbf{h}_k = e^{\boldsymbol{\Delta}_k \mathbf{A}} \mathbf{h}_{k-1} + \boldsymbol{\Delta}_k \mathbf{B}_k u_k, \quad y_k = \mathbf{C}_k^\top \mathbf{h}_k + D u_k, \quad (2)$$

scanned causally over the  $L$  steps and applied independently to each of the  $d_{\text{in}}$  feature dimensions: per dimension, the state is  $\mathbf{h}_k \in \mathbb{R}^N$ ,  $\mathbf{A} \in \mathbb{R}^N$  is diagonal, the input  $u_k$  is scalar, and  $(\boldsymbol{\Delta}_k \in \mathbb{R}, \mathbf{B}_k, \mathbf{C}_k \in \mathbb{R}^N)$  are produced from  $u$  by learned linear maps [8]. The last state, summed with a whole-window linear embedding  $\mathbf{W}_e \tilde{\mathbf{x}}^{(c)}$  (one per variate, as in [20]), forms the branch feature  $\mathbf{f}_{\text{time}} \in \mathbb{R}^{C \times d}$ . A head  $\phi_{\text{time}}$  maps features to a horizon correction:

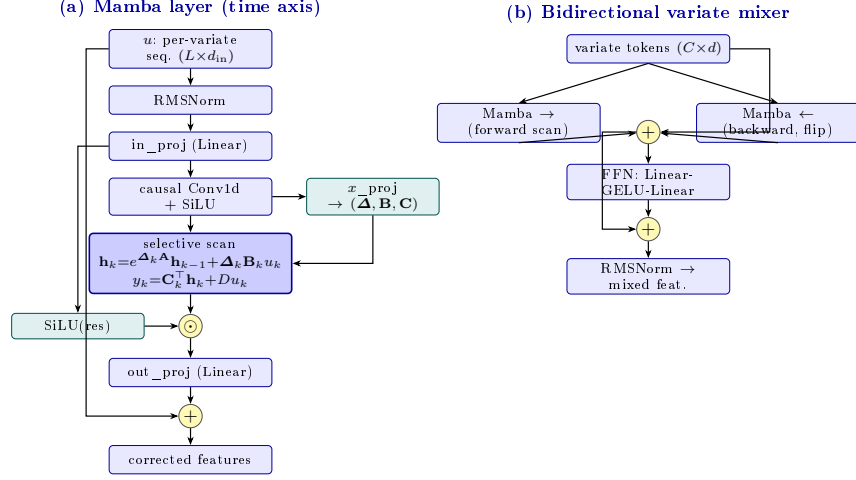
$$\mathbf{Y}_{\text{time}} = \mathbf{Y}_{\text{lin}} + \phi_{\text{time}}(\mathbf{f}_{\text{time}}), \quad \phi_{\text{time}} \equiv \mathbf{0} \text{ at init.} \quad (3)$$

Because  $\phi_{\text{time}}$  is zero-initialized, the branch’s output at initialization equals its (randomly initialized) DLinear-style backbone; the SSM learns only what the linear map cannot express. We run the recurrence in fp32 even under mixed precision, as the  $L$ -step accumulation overflows in half precision and silently destroys training.

### 3.3 Frequency Branch: Spectral Filtering and a Dormant Spectral Pathway

The frequency branch applies the real FFT along time,  $\mathbf{Z} = \text{rFFT}(\tilde{\mathbf{X}}) \in \mathbb{C}^{C \times F}$ ,  $F = \lfloor L/2 \rfloor + 1$ , optionally low-passed by zeroing the top fraction  $\rho$  of bins.

*Spectral encoder.* A learnable complex linear filter  $\mathbf{W} = \mathbf{W}_r + i\mathbf{W}_i \in \mathbb{C}^{F \times F}$  densely mixes bins,  $\mathbf{Z}' = \mathbf{Z}\mathbf{W}^\top$  along the frequency axis (a bidirectional Mamba over the bin sequence is an input-dependent alternative). The (real, imaginary) parts are projected to the branch feature  $\mathbf{f}_{\text{freq}} \in \mathbb{R}^{C \times d}$  and a head  $\phi_{\text{freq}}$  produces the spectral forecast.



**Fig. 2.** Internal structure of DD-Mamba’s two selective-state-space blocks (cf. the blue boxes in Fig. 1). **(a)** The causal Mamba layer used as the time-axis encoder (Eq. (2)): an input-dependent selective scan, gated by a SiLU branch and wrapped in a residual, with  $(\Delta, \mathbf{B}, \mathbf{C})$  predicted per step from the input. **(b)** The bidirectional mixer used over the *variate* dimension (Sect. 3.4): forward and backward scans of the  $C$  channel tokens summed in one residual stream, then a position-wise feed-forward block (the S-Mamba design [20]).

*FITS-style spectral pathway (optional, per dataset).* A complex linear map  $\mathbf{U} \in \mathbb{C}^{F' \times F}$  with  $F' = \lfloor (L+H)/2 \rfloor + 1$ , applied along the frequency axis ( $\mathbf{Z}\mathbf{U}^\top \in \mathbb{C}^{C \times F'}$ ), maps the (low-passed) window spectrum to that of a length- $(L+H)$  series, whose inverse transform’s last  $H$  samples are the branch’s linear spectral forecast:

$$\mathbf{Y}_{\text{freq}} = \underbrace{\text{irFFT}_{L+H}(\mathbf{Z}\mathbf{U}^\top)[L:]}_{\text{linear-in-frequency}} + \phi_{\text{freq}}(\mathbf{f}_{\text{freq}}), \quad \mathbf{U}, \phi_{\text{freq}} \equiv \mathbf{0} \text{ at init.} \quad (4)$$

This pathway uses the frequency interpolation idea of FITS [23], but is not a fitted FITS model:  $\mathbf{U}$  is *zero-initialized*, so the frequency branch is silent at initialization (the length- $L$  and length- $(L+H)$  frequency grids do not align, so no canonical identity init exists, and a random  $\mathbf{U}$  would inject noise during the highest-learning-rate epochs). With the time head, spectral map, and gate all zero or constant at init, the end-to-end map in the normalized space is the linear  $g_0 \cdot \mathbf{Y}_{\text{lin}}$ ; composed with RevIN this is the RLinear-class structure [11, 10] — linear in the normalized window, by construction not fitted. The spectral map is enabled per dataset (Table 2); where disabled, the frequency branch remains active through its encoder and forecast head.

### 3.4 Cross-Channel Mixing over Variates

Channel independence is a strong baseline [17, 25] but leaves accuracy on the table when variates are coupled [13, 20]. Inside each branch, before its head, we optionally run  $M$  layers of *bidirectional* Mamba over the  $C$  per-channel features, followed by a position-wise feed-forward block (Fig. 2b). Bidirectionality is essential: channel order carries no causal direction, so each layer sums a forward and a reversed scan in one residual stream. The mixer is a per-dataset choice:  $M=2$  at width 512 on electricity/traffic (its single most load-bearing component; Sect. 4.3),  $M=2$  at width 256 on weather,  $M=1$  on solar, and  $M=0$  on the few-channel ETT and exchange datasets, where it triples parameters without adding signal.

### 3.5 Forecast-Level Convex Fusion

Each branch emits a forecast and a feature. The model output is a gated convex combination

$$\mathbf{Y} = g \odot \mathbf{Y}_{\text{time}} + (1 - g) \odot \mathbf{Y}_{\text{freq}}, \quad g = \sigma(\psi([\mathbf{f}_{\text{time}}; \mathbf{f}_{\text{freq}}])) \in (0, 1)^{C \times 1}, \quad (5)$$

with  $\psi$  applied row-wise (per variate), followed by RevIN de-normalization. Since  $\mathbf{Y}$  lies elementwise between the branch forecasts, both domains retain an explicit prediction path, and the gate value reports each branch’s convex mixing weight — a lightweight proxy for its contribution, not a causal attribution. The constraint is structural, not behavioral:  $g$  is a sigmoid and can saturate near 0 or 1, so one branch may contribute very little — by design, as the ablations show the useful domain is dataset-dependent.  $\psi$  is zero-initialized with bias 2.2, so training starts at  $g \approx 0.9$ : the time branch is linear at initialization while the frequency branch is silent. An even 0.5/0.5 mix would therefore halve the initial linear forecast. This is an initialization convention; empirical gate distributions are not available in the present study.

### 3.6 Training

We minimize MSE with AdamW under a halving schedule ( $\eta_e = \eta_0 \cdot 2^{-(e-1)}$ ). All runs use 10 epochs with best-checkpoint early stopping. This schedule was held fixed across our runs but was not compared with cosine, warm-up, or longer-budget alternatives; schedule sensitivity therefore remains a potential confound.

## 4 Experiments

### 4.1 Setup

*Datasets and protocol.* We evaluate on nine standard benchmarks (Table 1) under the unified protocol of [13, 20]: look-back 96, horizons 96/192/336/720. ETT uses the canonical 12/4/4-month border split, the others chronological 0.7/0.1/0.2 splits; variates are standardized by training-set statistics; metrics are MSE and MAE.

**Table 1.** Benchmark datasets and where DD-Mamba places its capacity.

| Dataset      | Variates | Steps  | Granularity | Variate mixer   | Params     |
|--------------|----------|--------|-------------|-----------------|------------|
| ETTh1/ETTh2  | 7        | 17,420 | 1 hour      | off             | 0.25M      |
| ETTm1/ETTm2  | 7        | 69,680 | 15 min      | off             | 0.24–0.36M |
| Weather      | 21       | 52,696 | 10 min      | $M=2$ , $d=256$ | 5.8M       |
| Solar-Energy | 137      | 52,560 | 10 min      | $M=1$ , $d=128$ | 0.96M      |
| Electricity  | 321      | 26,304 | 1 hour      | $M=2$ , $d=512$ | 19.7M      |
| Traffic      | 862      | 17,544 | 1 hour      | $M=2$ , $d=512$ | 19.7M      |
| Exchange     | 8        | 7,588  | 1 day       | off             | 0.09M      |

*Baselines.* We compare against the strongest published results under this protocol: S-Mamba [20], iTransformer [13], PatchTST [17], DLinear [25], and TimesNet [21]. Numbers are quoted from [20] rather than reproduced: the protocol is identical, but training pipelines, seeds, and implementations differ across works, so gaps below the descriptive tolerance in Sect. 4.1 relative to quoted numbers should be read with caution (flagged in the text); a unified rerun is planned for the camera-ready appendix.

**Implementation and Reproducibility.** All experiments use PyTorch 2.11 (CUDA 13.0) on a single NVIDIA RTX 5090. Optimization is AdamW (grad clip 5.0), 10 epochs with the learning rate halved each epoch, early stopping on validation loss, and mixed precision with the selective scan forced to fp32 (Sect. 3.2). The decomposition kernel is 25; Mamba blocks use expansion 2 and conv width 4 throughout. Table 2 lists all per-dataset settings; configs and the ablation harness ship with the code. All main results are means over *three seeds* (2024–2026); the maximum per-entry std across the 36 dataset–horizon entries is 0.005 MSE, typically 0.001–0.003. The table records final empirical configurations, not a transferable capacity-selection rule: no common pre-registered search space or equalized tuning budget was used across all datasets. Consequently, capacity placement is reported as an empirical finding, and new datasets require validation-only selection of these switches.

*Statistical methodology.* We compare at two levels. *Within our framework* (ablations), every variant shares seeds with the reference, so we report *paired* per-seed differences with  $t$ -based 95% CIs ( $n=3$ ) and treat a component as supported only when its CI excludes zero. These intervals are exploratory:  $n=3$  limits power, and no family-wise correction is applied across ablations. *Against quoted baselines*, pairing is impossible, so 0.005 MSE is used only as a descriptive tolerance based on the largest standard deviation among our runs, not as a significance test. Our in-framework DLinear re-run shows why: through the identical pipeline on ETTh1 it averages 0.462/0.456 MSE/MAE versus its quoted 0.456/0.452 — a +0.006 implementation gap, the order of several margins in Table 3. Cross-paper comparisons therefore cannot support sub-0.01 distinctions; DD-Mamba’s

**Table 2.** Per-dataset hyperparameters.  $d$ : model width; time enc.: time-branch encoder (Mamba layers in parentheses);  $M$ : variate-mixer layers;  $N$ : SSM state size;  $\rho$ : spectral low-pass fraction (Sect. 3.3); spec. map: the *optional* FITS-style pathway of Eq. (4) — when “no”, the frequency branch remains fully active (rFFT, spectral encoder, forecast head) but carries no explicit linear pathway; IN: instance normalization (RevIN); do.: dropout; B: batch size; wd: weight decay. <sup>†</sup>Solar adopts IN = no from the ablation finding (Sect. 4.3); its main results were re-measured under this configuration across all horizons and seeds.

| Dataset     | $d$ | time enc. | $M$ | $N$ | $\rho$ | spec. map | IN              | lr                | B  | do./wd                |
|-------------|-----|-----------|-----|-----|--------|-----------|-----------------|-------------------|----|-----------------------|
| ETTh1       | 128 | Mamba (1) | 0   | 16  | 0.4    | yes       | yes             | $5 \cdot 10^{-4}$ | 32 | $0.2/10^{-4}$         |
| ETTh2       | 128 | Mamba (1) | 0   | 16  | 0.3    | yes       | yes             | $5 \cdot 10^{-4}$ | 32 | $0.3/5 \cdot 10^{-4}$ |
| ETTh1       | 128 | Mamba (2) | 0   | 16  | 0      | no        | yes             | $10^{-4}$         | 32 | $0.1/10^{-4}$         |
| ETTh2       | 128 | Mamba (1) | 0   | 16  | 0.2    | no        | yes             | $10^{-4}$         | 32 | $0.2/2 \cdot 10^{-4}$ |
| Weather     | 256 | Mamba (2) | 2   | 16  | 0      | yes       | yes             | $2 \cdot 10^{-4}$ | 32 | $0.1/10^{-4}$         |
| Solar       | 128 | Mamba (2) | 1   | 16  | 0      | no        | no <sup>†</sup> | $5 \cdot 10^{-4}$ | 16 | $0.1/10^{-4}$         |
| Electricity | 512 | MLP       | 2   | 32  | 0      | no        | yes             | $5 \cdot 10^{-4}$ | 16 | $0.1/10^{-4}$         |
| Traffic     | 512 | MLP       | 2   | 32  | 0      | no        | yes             | $10^{-3}$         | 16 | $0.1/10^{-4}$         |
| Exchange    | 64  | Mamba (1) | 0   | 8   | 0.5    | no        | yes             | $10^{-4}$         | 32 | $0.3/5 \cdot 10^{-4}$ |

**Table 3.** Long-term forecasting results averaged over horizons  $H \in \{96, 192, 336, 720\}$  with look-back  $L=96$  (MSE/MAE; lower is better). DD-Mamba entries are means over three seeds (Table 4 reports per-horizon std). Baselines quoted from [20]. **Bold** marks the best result *and* every result within the descriptive tolerance ( $< 0.005$ , Sect. 4.1) of it; this formatting is not a cross-paper significance claim; underline marks the next-best result.

| Dataset     | DD-Mamba (ours)    | S-Mamba            | iTransformer       | PatchTST           | DLinear            | TimesNet    |
|-------------|--------------------|--------------------|--------------------|--------------------|--------------------|-------------|
| ETTh1       | <b>0.439/0.434</b> | 0.455/0.450        | <u>0.454/0.447</u> | 0.469/0.454        | 0.456/0.452        | 0.458/0.450 |
| ETTh2       | <b>0.374/0.401</b> | <u>0.381/0.405</u> | <u>0.383/0.407</u> | <u>0.387/0.407</u> | 0.559/0.515        | 0.414/0.427 |
| ETTh1       | <b>0.386/0.398</b> | <u>0.398/0.405</u> | 0.407/0.410        | <b>0.387/0.400</b> | 0.403/0.407        | 0.400/0.406 |
| ETTh2       | <b>0.280/0.325</b> | <u>0.288/0.332</u> | <u>0.288/0.332</u> | <b>0.281/0.326</b> | 0.350/0.401        | 0.291/0.333 |
| Weather     | <b>0.248/0.276</b> | <b>0.251/0.276</b> | <u>0.258/0.278</u> | <u>0.259/0.281</u> | 0.265/0.317        | 0.259/0.287 |
| Electricity | <b>0.168/0.263</b> | <b>0.170/0.265</b> | <u>0.178/0.270</u> | 0.205/0.290        | 0.212/0.300        | 0.192/0.295 |
| Traffic     | 0.438/0.281        | <b>0.414/0.276</b> | <u>0.428/0.282</u> | 0.481/0.304        | 0.625/0.383        | 0.620/0.336 |
| Exchange    | 0.363/0.405        | 0.367/0.408        | <u>0.360/0.403</u> | <b>0.367/0.404</b> | <b>0.354/0.414</b> | 0.416/0.443 |
| Solar       | <b>0.199/0.260</b> | 0.240/0.273        | <u>0.233/0.262</u> | 0.270/0.307        | 0.330/0.401        | 0.301/0.319 |

margin over the in-framework DLinear (0.439 vs. 0.462) is, by contrast, a same-pipeline comparison.

## 4.2 Main Results

Against the quoted results and under the stated descriptive tolerance, DD-Mamba shows average-MSE gaps larger than 0.005 on Solar, ETTh1, and ETTh2 (Table 3; per-horizon values in Table 4), although only the Solar (0.034) and ETTh1 (0.015) margins also exceed the wider 0.01 cross-paper caution of Sect. 4.1; ETTh2’s 0.007 lies between the two thresholds. The largest margin is on Solar (0.199 vs. iTransformer’s 0.233;  $-17\%$  vs. S-Mamba, with the normalization-free configuration of Sect. 4.3). On ETTh1 and ETTh2 it improves over the quoted S-Mamba result but ties PatchTST, and on Weather and Electricity the gap is

**Table 4.** DD-Mamba per-horizon results (mean  $\pm$  std over three seeds), look-back  $L=96$ .

| Dataset     |     | $H=96$           | $H=192$          | $H=336$          | $H=720$          | Avg   |
|-------------|-----|------------------|------------------|------------------|------------------|-------|
| ETTh1       | MSE | 0.380 $\pm$ .001 | 0.431 $\pm$ .002 | 0.474 $\pm$ .004 | 0.470 $\pm$ .003 | 0.439 |
|             | MAE | 0.396 $\pm$ .001 | 0.424 $\pm$ .001 | 0.447 $\pm$ .003 | 0.468 $\pm$ .001 | 0.434 |
| ETTh2       | MSE | 0.293 $\pm$ .001 | 0.368 $\pm$ .003 | 0.414 $\pm$ .001 | 0.421 $\pm$ .001 | 0.374 |
|             | MAE | 0.344 $\pm$ .002 | 0.391 $\pm$ .000 | 0.426 $\pm$ .000 | 0.441 $\pm$ .001 | 0.401 |
| ETTM1       | MSE | 0.320 $\pm$ .001 | 0.366 $\pm$ .002 | 0.398 $\pm$ .001 | 0.462 $\pm$ .001 | 0.386 |
|             | MAE | 0.359 $\pm$ .001 | 0.384 $\pm$ .001 | 0.406 $\pm$ .001 | 0.443 $\pm$ .001 | 0.398 |
| ETTM2       | MSE | 0.177 $\pm$ .000 | 0.242 $\pm$ .002 | 0.302 $\pm$ .000 | 0.401 $\pm$ .003 | 0.280 |
|             | MAE | 0.260 $\pm$ .001 | 0.303 $\pm$ .001 | 0.340 $\pm$ .000 | 0.398 $\pm$ .002 | 0.325 |
| Weather     | MSE | 0.162 $\pm$ .000 | 0.212 $\pm$ .000 | 0.269 $\pm$ .001 | 0.352 $\pm$ .003 | 0.248 |
|             | MAE | 0.207 $\pm$ .001 | 0.255 $\pm$ .000 | 0.295 $\pm$ .002 | 0.349 $\pm$ .002 | 0.276 |
| Electricity | MSE | 0.140 $\pm$ .000 | 0.157 $\pm$ .001 | 0.174 $\pm$ .001 | 0.200 $\pm$ .004 | 0.168 |
|             | MAE | 0.235 $\pm$ .000 | 0.252 $\pm$ .001 | 0.271 $\pm$ .001 | 0.296 $\pm$ .004 | 0.263 |
| Traffic     | MSE | 0.402 $\pm$ .002 | 0.424 $\pm$ .001 | 0.441 $\pm$ .001 | 0.485 $\pm$ .003 | 0.438 |
|             | MAE | 0.266 $\pm$ .001 | 0.275 $\pm$ .001 | 0.283 $\pm$ .000 | 0.301 $\pm$ .000 | 0.281 |
| Exchange    | MSE | 0.086 $\pm$ .000 | 0.179 $\pm$ .001 | 0.330 $\pm$ .002 | 0.857 $\pm$ .001 | 0.363 |
|             | MAE | 0.205 $\pm$ .000 | 0.301 $\pm$ .000 | 0.416 $\pm$ .002 | 0.699 $\pm$ .000 | 0.405 |
| Solar       | MSE | 0.180 $\pm$ .004 | 0.195 $\pm$ .005 | 0.207 $\pm$ .001 | 0.212 $\pm$ .005 | 0.199 |
|             | MAE | 0.237 $\pm$ .003 | 0.259 $\pm$ .003 | 0.270 $\pm$ .001 | 0.274 $\pm$ .005 | 0.260 |

below the tolerance. It trails DLinear on Exchange (by 0.009, above tolerance) and S-Mamba on Traffic (by 0.024), which we *hypothesize* reflects the weekly cycle exceeding the look-back. As all comparisons are to numbers *reported* in [20], we phrase the claim conservatively: DD-Mamba is competitive with results reported under this protocol, its clearest margins on Solar and ETTh1, where the linear backbone and capacity restraint matter most and DD-Mamba uses  $\approx 0.25\text{M}$  parameters — roughly an order of magnitude fewer than variate-attention models configured for these datasets.

### 4.3 Component Ablations

Our ablation harness flips exactly one switch of a dataset’s full configuration, holding split, schedule, seeds, and every other hyperparameter fixed. We report its complete output on three datasets under their *final released configurations*, three seeds each at  $H=96$ : ETTh1 and Solar (Table 5) and Weather (Table 7), with the paired methodology of Sect. 4.1. Two consistency checks pass: the Solar “full” row reproduces the main-table Solar  $H=96$  entry (0.180 $\pm$ .004), and its w/o-instance-norm variant coincides with the full (RevIN-off) configuration, so that cell is omitted.

*Primary finding: capacity choices can help or hurt.* The most robust result in our studies is two-sided and spans datasets. Adding the variate mixer on ETTh1 *hurts* (+0.0077\*) while removing it on Weather also hurts (+0.0104\*); normalization shows the same two-sided pattern (removing it hurts Weather +0.0161\*, while disabling it on Solar improved the re-measured main results from 0.228 to 0.199). Zero initialization is a *null* result — no significant difference on either dataset (−0.0012 ETTh1, −0.0046 Solar; CIs include zero) — so we make no accuracy claim for it, only the interpretable linear start (Sect. 3.2). The DLinear

**Table 5.** Component ablations on ETTh1 and Solar-Energy ( $H=96$ , three seeds, final configurations): paired per-seed  $\Delta$ MSE to the full model with  $t$ -based 95% CIs; \* CI excludes zero. The spectral map is enabled on ETTh1 (hence *removed*) and disabled on Solar (hence *added*); the variate mixer is off on ETTh1 (*added*) and on for Solar (*removed*).

| Variant                         | ETTh1: $\Delta$ [95% CI]  | Solar: $\Delta$ [95% CI]  |
|---------------------------------|---------------------------|---------------------------|
| full (reference MSE)            | 0.3802 $\pm$ .001         | 0.1802 $\pm$ .004         |
| time branch only (freq removed) | +0.0034 [−0.009, +0.016]  | +0.0114 [+0.001, +0.021]* |
| freq branch only (time removed) | −0.0007 [−0.004, +0.003]  | +0.0051 [−0.013, +0.023]  |
| sum fusion                      | −0.0026 [−0.007, +0.002]  | +0.0066 [−0.005, +0.018]  |
| w/o instance norm               | +0.0061 [−0.001, +0.013]  | —                         |
| w/o linear backbone             | −0.0031 [−0.009, +0.003]  | +0.0081 [−0.019, +0.035]  |
| random init (vs. zero-init)     | −0.0012 [−0.005, +0.002]  | −0.0046 [−0.022, +0.013]  |
| spectral map removed / added    | +0.0042 [−0.018, +0.026]  | +0.0088 [−0.002, +0.019]  |
| variate mixer added / removed   | +0.0077 [+0.004, +0.012]* | +0.0119 [−0.010, +0.034]  |
| Mamba→MLP over time             | −0.0022 [−0.007, +0.003]  | +0.0089 [−0.002, +0.020]  |
| freq Mamba encoder              | −0.0024 [−0.011, +0.006]  | −0.0117 [−0.044, +0.020]  |

backbone and the learned gate versus averaging are within CIs everywhere; on Weather (Table 7) three components have unadjusted paired CIs excluding zero (normalization, the variate mixer, and the temporal Mamba, +0.0027, positive on all seeds). The recurring pattern — component value is dataset-dependent, in *both* directions — is the argument for per-dataset switches over universal defaults, and it does not depend on any single dataset.

*Branch matrix: where each domain matters.* Because a single Solar cell cannot carry a dual-domain conclusion, we ran the full branch-ablation matrix — full / time-only / freq-only on all nine datasets at  $H=96$  under paired seeds, plus an ETTh1 horizon sweep (Table 6). Every full-model mean reproduces the corresponding main-table entry, and the Solar cell replicates the earlier ablation (+0.0126 vs. +0.0114, overlapping CIs). Removing the frequency branch significantly degrades *four* of nine datasets — Solar (+0.0126\*), ETTm1 (+0.0073\*), Electricity (+0.0019\*), and Exchange (+0.0015\*) — and removing the time branch significantly degrades four (ETTh1 +0.0196\*, ETTm2 +0.0060\*, Exchange +0.0058\*, Electricity +0.0017\*); on ETTm1, Electricity, and Exchange *both* branches are individually necessary. On ETTh1, ETTh2, Weather, and Traffic no branch-removal effect is detected, and the ETTh1 horizon sweep shows this redundancy persists at  $H=192/336/720$  — it is not an artifact of the short horizon. The dual-domain claim is therefore supported, but dataset-conditional: the frequency path earns its place on four benchmarks and is a no-cost redundancy on the rest.

*Negative results that shaped the design.* Deepening the traffic mixer from two to four layers (19.7  $\rightarrow$  37.9M) gave no gain, so the release keeps two; enabling the mixer on ETT (8.4K training windows) tripled parameters and was associated with premature early stopping and worse held-out error, so the released ETT configurations disable it.

**Table 6.** Branch-ablation matrix: paired per-seed  $\Delta$ MSE versus the full model when one branch is removed (three seeds;  $t$ -based 95% CIs; \* CI excludes zero). All nine datasets at  $H=96$ , plus an ETTh1 horizon sweep. Full-model means match Table 4.

| Dataset     | $H$ | freq removed: $\Delta$ [95% CI] | time removed: $\Delta$ [95% CI] |
|-------------|-----|---------------------------------|---------------------------------|
| ETTh1       | 96  | +0.0034 [−0.0089, +0.0157]      | −0.0007 [−0.0039, +0.0025]      |
| ETTh2       | 96  | +0.0006 [−0.0203, +0.0215]      | −0.0031 [−0.0064, +0.0003]      |
| ETTh1       | 96  | +0.0073 [+0.0029, +0.0118]*     | +0.0196 [+0.0179, +0.0213]*     |
| ETTh2       | 96  | +0.0020 [−0.0004, +0.0045]      | +0.0060 [+0.0049, +0.0071]*     |
| Weather     | 96  | +0.0008 [−0.0020, +0.0036]      | +0.0006 [−0.0039, +0.0051]      |
| Solar       | 96  | +0.0126 [+0.0077, +0.0175]*     | +0.0052 [−0.0135, +0.0238]      |
| Electricity | 96  | +0.0019 [+0.0006, +0.0032]*     | +0.0017 [+0.0014, +0.0020]*     |
| Traffic     | 96  | −0.0015 [−0.0092, +0.0061]      | +0.0013 [−0.0052, +0.0078]      |
| Exchange    | 96  | +0.0015 [+0.0001, +0.0029]*     | +0.0058 [+0.0047, +0.0069]*     |
| ETTh1       | 192 | +0.0005 [−0.0140, +0.0150]      | −0.0001 [−0.0066, +0.0065]      |
| ETTh1       | 336 | −0.0047 [−0.0248, +0.0154]      | −0.0003 [−0.0142, +0.0137]      |
| ETTh1       | 720 | +0.0096 [−0.0387, +0.0579]      | +0.0176 [−0.0019, +0.0372]      |

**Table 7.** Component ablation on Weather ( $H=96$ ; MSE mean  $\pm$  std over three seeds).  $\Delta$  is the *paired* per-seed difference to the full model with a  $t$ -based 95% confidence interval ( $n=3$ ); \* CI excludes zero.

| Variant                           | MSE                | paired $\Delta$ [95% CI]    | Component isolated                 |
|-----------------------------------|--------------------|-----------------------------|------------------------------------|
| full                              | 0.1620 $\pm$ .0003 | —                           | reference                          |
| w/o instance norm                 | 0.1781 $\pm$ .0035 | +0.0161 [+0.0053, +0.0269]* | RevIN                              |
| w/o channel mixer                 | 0.1724 $\pm$ .0021 | +0.0104 [+0.0037, +0.0172]* | cross-variate Mamba mixing         |
| Mamba $\rightarrow$ MLP over time | 0.1647 $\pm$ .0005 | +0.0027 [+0.0020, +0.0034]* | selective scan (time)              |
| w/o linear backbone               | 0.1631 $\pm$ .0006 | +0.0011 [−0.0014, +0.0036]  | DLinear anchor (time)              |
| time branch only                  | 0.1628 $\pm$ .0011 | +0.0007 [−0.0020, +0.0035]  | frequency branch removed           |
| w/o spectral map                  | 0.1627 $\pm$ .0016 | +0.0007 [−0.0041, +0.0055]  | FITS-style pathway                 |
| freq branch only                  | 0.1626 $\pm$ .0014 | +0.0006 [−0.0040, +0.0052]  | time branch removed                |
| freq Mamba encoder                | 0.1626 $\pm$ .0016 | +0.0006 [−0.0044, +0.0055]  | spectral encoder type              |
| sum fusion                        | 0.1622 $\pm$ .0007 | +0.0001 [−0.0020, +0.0023]  | learned gate $\rightarrow$ average |

#### 4.4 Analysis

*Where each domain wins.* The matrix (Table 6) separates three regimes. On Solar the frequency path dominates: removing it costs +0.0126\* while removing the time branch is within noise — yet because Solar’s 10-minute daily cycle spans  $144 > L=96$  samples, the spectral path cannot observe a complete daily period, so its benefit must come from shorter-scale spectral structure. On ETTh1, Electricity, and Exchange the branches are complementary — removing *either* one significantly degrades the forecast. On ETTh1 (at all four horizons), ETTh2, Weather, and Traffic the branches are interchangeable, and the convex gate makes carrying the unused path harmless. This heterogeneity is the dual-domain analogue of the capacity-placement finding: which domain a dataset needs is itself dataset-dependent.

*Complexity accounting.* Table 8 reports model-only, hardware-independent estimates rather than comparative runtime evidence. Parameter count is governed by model *width*, not channel count (electricity and traffic share 19.7M despite a  $2.7\times$  gap in variates), so model size is decoupled from data dimensionality; com-

**Table 8.** Model-only complexity profile of DD-Mamba per dataset (look-back 96, horizon 96, batch 1). GMACs: multiply-accumulates per forecast (**Linear** + **Conv1d** + the selective-scan recurrence); act. mem.: forward activation footprint (fp32). All three columns are hardware-independent estimates; no wall-clock or matched-baseline runtime is claimed.

| Dataset     | Variates | Params (M) | GMACs | Act. mem. (MB) |
|-------------|----------|------------|-------|----------------|
| Exchange    | 8        | 0.09       | 0.02  | 2.3            |
| ETTh1       | 7        | 0.25       | 0.08  | 4.0            |
| Weather     | 21       | 5.77       | 1.90  | 46.6           |
| Solar       | 137      | 0.96       | 3.26  | 150.4          |
| Electricity | 321      | 19.70      | 22.4  | 254.3          |
| Traffic     | 862      | 19.70      | 60.2  | 682.7          |

pute and activation memory instead scale with the channel count through the variate mixer (traffic 60 GMACs vs. electricity’s 22 at equal width, while ETT and exchange sit two orders of magnitude below), keeping the capacity restraint on those datasets inexpensive. Wall-clock throughput and matched-baseline runtime remain unmeasured.

## 5 Conclusion

DD-Mamba combines a DLinear-style temporal backbone, a selective-state-space correction, an optional frequency pathway, and dataset-dependent cross-variate mixing. Against quoted results under the look-back-96 protocol, it shows average-MSE gaps above the descriptive tolerance on Solar, ETTh1, and ETTh2 (only Solar and ETTh1 also clearing the wider 0.01 cross-paper caution), lies within that tolerance of the strongest reported result on four datasets, and trails on Exchange and Traffic. The ablations support two conclusions. First, *placement*: the variate mixer helps Weather yet hurts ETTh1 and normalization helps Weather yet hurts Solar, each direction backed by a paired CI excluding zero on a different dataset. Second, the nine-dataset branch matrix locates the dual-domain benefit: the frequency branch is individually significant on Solar, ETTm1, Electricity, and Exchange, both branches are jointly necessary on three of those, and the branches are redundant on ETTh1 (across all four horizons), ETTh2, Weather, and Traffic. Zero initialization is accuracy-neutral and retained only for its controlled linear starting point. These findings establish conditional component placement: dual-domain value is real on four of nine benchmarks and structurally harmless on the rest, not a universal gain.

The evidence remains limited in several ways. Baselines other than DLinear on ETTh1 are quoted rather than rerun in one pipeline; the branch matrix covers all datasets only at  $H=96$  (the horizon sweep is ETTh1-only), the other component ablations only  $H=96$ , all at three seeds without family-wise correction; schedule sensitivity and head-norm dynamics are not measured; and the

complexity table lacks wall-clock and matched-baseline runtime. The fixed 96-step look-back also leaves the Traffic weekly-cycle hypothesis untested, while Exchange at  $H=720$  remains a notable failure case. Explicit forecasting of time-varying dispersion is left as future work rather than claimed as part of the evaluated model.

*Reproducibility.* Code, configurations, and the ablation harness are supplementary material, to be released publicly upon acceptance.

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